What I need you to measure

Updated 2026-07-31. For Mridul.

Every number here is something the CAD is currently **guessing** - typed from a datasheet or a forum post, never taken off your actual parts. Where a guess is wrong, the printed part won't fit the real one, and you only find out after paying for the print.

26 measurements. About 45 minutes.

Each one is written twice:

> [OK] **In plain words** - what to actually do, no jargon **Technical name** - the proper term, so you recognise it in datasheets, shops and my messages

Copy-paste answer sheet at the very bottom. Fill it in, send it back. Write ? for anything you can't measure - a ? is far more useful to me than a guess.

PART 0 - Words you'll keep seeing

Learn these six and everything below reads easily.

Word	What it actually means
Diameter	The width straight across a circle, through the middle. Not around it.
Bore	A hole. "Bore diameter" = how wide the hole is.
Pitch	The gap from the centre of one thing to the centre of the next - holes, pins, teeth. Centre-to-centre, **never** edge-to-edge.
Boss	A raised bump or collar sticking out of a surface.
Ear / lug / tab	A flat sticking-out tab with a screw hole in it.
Flat-to-flat	The width across the two flat sides of a rod that isn't fully round.

Rule of thumb: anything called "diameter" or "width" is measured with the **big jaws**. Anything called "bore" or "hole" is measured with the **small pointy jaws** on top.

PART 0.5 - Using the calipers (30 seconds)

```
        +-- BIG jaws: measure the OUTSIDE of things
        ?
   ---+ +------------------------------
      | |    [ 12.47 ]  <- screen
   ---+ +------------------------------
        ?
        +-- small pointy jaws on TOP: measure the INSIDE of holes
```

- **mm/inch button** - make sure it says **mm**. If a big object reads 0.5 or 1.3, you're in inches.
- **ZERO button** - close the jaws fully, press ZERO, *every session*. Otherwise every reading is

off by the same amount.

- **Close the jaws gently.** Plastic squashes and you'll read too small. Stop when it stops sliding.
- **Write 2 decimal places.** "12" is useless. "12.47" is what I need.

PART 1 - THE MOTOR * MOST IMPORTANT

The silver can with the blue plastic base and 5 coloured wires.

Why this one matters most: the motor's height sets how high the base plate sits -> which sets how high the cam sits -> which sets how long the linkages are. Get the height wrong by 2mm and **all six braille dots stick up permanently** - every dot reads as "on" and the display is useless.

The CAD says 19mm tall. Nobody has ever checked.

```
                    shaft (the thin metal rod that spins)
                         |
                    +----+----+
              +-----+  boss   +-----+   <- boss = raised collar round the shaft
   ear --->  ?      +---------+      ?
```

```
 ()                          ()  <--- ear = flat tab with a hole
 ?                          ?
  +-----------------------+  <--- THIS flat face is the "mounting face"
  +-----------------------+   ?
  |                       |   | can height
  |      silver can       |   |
  |                       |   ?
  +-----------------------+
```

M1

[OK] **How wide is the round silver part?** Measure straight across it at its fattest point. **Can outer diameter** *(guessed: 28mm)*

*M2 ** THE CRITICAL ONE*

[OK] **How tall is the motor, from the table up to the flat face where the two tabs are?** Stand it silver-can-down on a table. Measure up to that flat face - **not** to the top of the spinning rod. Just the body. **Can height, base to mounting face** *(guessed: 19mm)*

M3

[OK] **The spinning rod is NOT in the middle of the can. How far off-centre is it?** Look down from above. Measure from the middle of the rod to the middle of the round can. **Shaft-to-body-centre offset** *(guessed: 8mm)* *Too fiddly? Instead measure rod-centre to the left edge of the can, and rod-centre to the right edge. Give me both and I'll work it out.*

M4

[OK] **How thick is the spinning rod at its widest?** It's not perfectly round - it has two flat sides shaved off. Measure the **round** way, the fat direction. **Shaft diameter** *(guessed: 5.0mm)*

M5

[OK] **Now measure the rod the narrow way - across the two flat sides.** Smaller number. **Shaft flat-to-flat / across-flats** *(guessed: 3.0mm)* *That shape is called a **double-D**. It's what stops the cam spinning loose on the rod. Both M4 and M5 must be right or the cam either won't go on, or will wobble.*

M6

[OK] **How far does the rod stick up above that flat face?** From the flat face (same one as M2) to the very tip. **Shaft length above the mounting face** *(guessed: 10mm)*

M7

[OK] **Is there a small raised ring where the rod comes out of the body? How wide and how tall? If there's no raised ring, just write "none". Shaft boss diameter and height** *(guessed: 9mm wide; height never guessed at all)*

M8

[OK] **How far apart are the two screw holes in the tabs?** Measure from the middle of one hole to the middle of the other. **Mounting-hole pitch (centre-to-centre)** *(guessed: 35mm)* *Middle-to-middle is awkward. Easier: measure from the **far outer edge** of one hole to the **far outer edge** of the other, tell me that, and I'll subtract the hole width.*

M9

[OK] **How wide is one of those screw holes?** Use the small pointy jaws inside it. **Mounting-hole bore diameter** *(guessed: 4.2mm)*

M10

[OK] **How wide is the flat tab, and how thick is the metal? Mounting ear width and thickness** *(never measured)*

PART 2 - THE POSITION SENSOR

The small **blue board** with a tiny black 3-legged part on one edge, a blue adjustment screw, and pins marked AO DO GND VCC.

[!] Bad news first

The blue board will not fit inside the machine. It's roughly 15x11mm plus tall pins. The space inside the base plate is 5mm thick. There's no version where the whole board fits.

What has to happen: the little black 3-legged part on the edge *is* the actual sensor - everything else on that board is supporting circuitry. You'll need to **unsolder that black part** and mount just it, running three wires back.

Three solder joints. Fiddly, but genuinely the only option: it has to sit within a couple of millimetres of a magnet on the spinning disc, and the motor body fills everything else nearby.

You do NOT need to do this for the demo. On a breadboard the blue board works fine as-is.

M11 - the black 3-legged part

You can measure it while it's still soldered on.

```
+---------+   ?
|  flat   |   | HEIGHT (M11c)
|  face   |   |
+-+--+--+-+   ?
  |  |  |     <- 3 legs
<--------->
  WIDTH (M11a)

from the side:  <->  THICKNESS (M11b)
```

[OK] **M11a - How wide is the black part across its flat face?** Sensor body width *(guessed: 4.1mm)*

[OK] **M11b - How thick is it front-to-back?** One side is flat, the other bulges. [!] **This is the one that can break the design** - I've allowed 1.6mm. If yours is fatter, the CAD will refuse to build and say so. Send me the number and I'll rework it. Sensor body thickness *(guessed: 1.6mm)*

[OK] **M11c - How tall is just the black block?** Not counting the legs. Sensor body height *(guessed: 3.1mm)*

[OK] **M11d - How far apart are the legs?** Middle of one leg to middle of the next. Lead pitch *(guessed: 1.27mm)*

PART 3 - THE POWER SOCKET

The small **yellow/black block** with a round socket at one end and two screw terminals at the other. Where the 5V adapter plugs in.

Every single dimension for this is a placeholder I invented. The printed holder meant to support it is entirely fictional until you measure this.

```
screw terminals             round socket
      |                          |
+-----?--------------------------?-----+  ?
|              yellow body             |  | HEIGHT (M13)
+--------------------------------------+  ?
<------------  LENGTH (M12)  ----------->

from the front:  <- WIDTH (M14) ->
```

[OK] **M12 - How long is the block end to end?** Terminals to socket face. Jack body length *(placeholder: 12mm)*

[OK] **M13 - How tall is it?** Jack body height *(placeholder: 11mm)*

[OK] **M14 - How wide is it?** Jack body width *(placeholder: 9.5mm)*

[OK] **M15 - How wide is the round metal tube the plug pushes into?** Measure across the tube at the front face. Barrel outer diameter *(placeholder: 11.5mm)*

[OK] **M16 - Does it have a screw thread and a ring nut** for clamping it to a panel? Yours probably doesn't. **Just answer YES or NO.** If yes, how wide is the threaded part? Panel-mount thread present? Thread diameter?

[!!] M17 - POLARITY. Do this one FIRST. It is not a measurement.

This is the single most likely way to destroy your ?350 ESP32. There is no protection against getting it backwards anywhere in this circuit - plug it in wrong and the board dies instantly and silently.

[OK] **Which of the two screw terminals is the positive one?** Supply polarity

Multimeter on **DC volts**, adapter plugged into the wall, **nothing else connected**:

1. Black probe on one screw terminal, red probe on the other.
2. Screen shows **+5V** -> the terminal under the **red** probe is **positive**.
3. Screen shows **?5V** (with a minus) -> it's the other way round.
4. Mark the positive one with nail polish or a marker.

Answer: is positive the terminal nearer the round socket, or further from it? Do not skip this. Do not guess from wire colours.

PART 4 - THE ESP32 BOARD

The black board with the silver square shield and USB-C.

*M18 ** FREE. TEN SECONDS. DO IT NOW.*

[OK] **Count the metal pins along ONE long edge.** Just count them. Pin count per row

- **15 per side** (30 total) -> you have the "30-pin" board
- **19 per side** (38 total) -> the "38-pin" board

Why this matters: the pod has two printed slots the board's pins drop into. The two versions have their pin rows **2.5mm further apart**. Every document in this project says you have the 30-pin board, but the CAD was built for the 38-pin one. **One of them is wrong - and if it's the CAD, your ESP32 will not go into the pod at all.**

[OK] **M19 - How long is the board?** The long way. Board length *(guessed: 51.5mm)*

[OK] **M20 - How wide is the board?** The short way. Board width *(guessed: 28.0mm)*

[OK] **M21 - How far apart are the two rows of pins?** Middle of the left row to middle of the right row. Header row pitch *(guessed: 25.4mm)* *Easier: measure from the **outer edge** of the left pins to the **outer edge** of the right pins and tell me that instead.*

[OK] **M22 - The silver USB socket on the board:** how wide (a), how tall (b), and how far does it stand up off the board surface (c)? USB connector width / height / height above PCB

[OK] **M23 - Your USB cable's plug:** not the metal bit - the **plastic body behind it**, the part your fingers hold. How wide (a) and how tall (b)? Connector overmould width and height *Why: the socket sits ~2mm inside the pod, so the plug body has to pass through the hole in the wall too. The old hole was 10x7mm and would probably have blocked your cable. The service opening is now fixed at 14x9mm. M23 is optional unless your cable body is larger than that.*

PART 5 - THE MOTOR DRIVER (blue board, 4 red LEDs)

[OK] **M24 - How long and how wide is the blue board?** PCB length x width

[OK] **M25 - How tall is it once the wires are soldered flat against it?** Assembled stack height *Measure **after** soldering, or press the wires flat and estimate. Do **not** measure it with the plug-in jumper wires standing upright - that makes it ~20mm and the space is 16mm. That's exactly why the wires must be soldered flat.*

[OK] **M26 - Which part on the board stands up tallest, and how tall is it** from the board surface? Tallest component height

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PART 6 - Quick confirmations (10 seconds each)

[OK] **M27 - One of your silver disc magnets: how wide and how thick?** Magnet diameter x thickness *(expect 8mm x 1mm)*

[OK] **M28 - One tiny black push-button: how wide is the square body, and how tall is it including the round nub on top?** Tactile switch body and total height *(expect 6 x 6 x 5mm)*

[!] Shopping correction that came out of this work

Your BOM used to say buy a **3mm x 2mm** homing magnet.

Buy 3mm x 1mm instead.

The disc it glues into is only 2mm thick. A 2mm-deep pocket cut clean through it, punching a 3.2mm hole across three of the six cam tracks - and a linkage foot crossing that hole would drop in and jam the whole mechanism. A 1mm magnet leaves 0.8mm of material and works just as well, because it still sits flush against the underside.

Already ordered 3x2mm? They'll be useful elsewhere. Just buy 3x1mm too. ~?120 for ten.

? COPY-PASTE ANSWER SHEET

Copy all of this, fill in the numbers, send it back. ? is fine and useful.

```
=== MOTOR (the silver can) ===
M1  width across the silver can            = ____ mm
M2  height, table to the flat tab face     = ____ mm   *** MOST IMPORTANT ***
M3  how far the rod is off-centre          = ____ mm
M4  rod thickness, the fat way             = ____ mm
M5  rod thickness, across the flats        = ____ mm
M6  how far the rod sticks up              = ____ mm
M7a width of the raised ring round the rod = ____ mm   (or "none")
M7b height of that ring                    = ____ mm   (or "none")
M8  gap between the 2 screw holes, mid-mid = ____ mm
M9  width of one screw hole                = ____ mm
M10a width of the flat tab                 = ____ mm
M10b thickness of the tab metal            = ____ mm

=== POSITION SENSOR (the black 3-legged part) ===
M11a width across its flat face            = ____ mm
M11b thickness front-to-back               = ____ mm   *** can break the design ***
M11c height of the black block only        = ____ mm
M11d gap between legs, middle to middle    = ____ mm

=== POWER SOCKET (yellow block) ===
M12 length end to end                      = ____ mm
M13 height                                 = ____ mm
M14 width                                  = ____ mm
M15 width of the round tube at the front   = ____ mm
M16 screw thread + ring nut?               = YES / NO   (thread width if yes: ____ mm)
M17 POSITIVE terminal is                   = NEARER the round socket / FURTHER from it
                                             *** do this before powering anything ***

=== ESP32 BOARD ===
M18 pins along ONE long edge               = ____ pins  *** just count them ***
M19 board length                           = ____ mm
M20 board width                            = ____ mm
M21 gap between the 2 pin rows, mid-mid    = ____ mm
    (or outer-edge to outer-edge:            ____ mm)
M22a USB socket width                      = ____ mm
M22b USB socket height                     = ____ mm
M22c how far it stands off the board        = ____ mm
M23a cable plug plastic body width         = ____ mm
M23b cable plug plastic body height        = ____ mm

=== MOTOR DRIVER (blue board) ===
M24a board length                          = ____ mm
M24b board width                           = ____ mm
M25  height with wires soldered FLAT       = ____ mm
M26a tallest part on it is                 = ____________
M26b its height above the board            = ____ mm

=== QUICK CONFIRMATIONS ===
M27a magnet width                          = ____ mm   (expect 8)
M27b magnet thickness                      = ____ mm   (expect 1)
M28a push-button square body               = ____ mm   (expect 6)
M28b push-button total height              = ____ mm   (expect 5)
```

What happens when you send these back

I re-derive the **whole vertical stack in one pass** - motor height -> base plate -> cam -> linkage length - rather than patching one number at a time. That's deliberate: these all depend on each other, so fixing them individually just moves the error somewhere else.

Four CAD problems are waiting on these numbers and cannot be fixed without them:

Waiting on	What it fixes
M2, M6, M7	The cam sits 2mm too high, so every dot is permanently raised and unreadable
M2	A second, separate 2mm error - the mid-plate rests *on top of* its ledge, not level with it, so the motor face lands 2mm higher than the box expects
M11b	Whether the sensor actually fits the pocket I built for it
M12-M16	The power-socket holder, which is currently 100% invented

Waiting on	What it fixes
M18, M21	Whether your ESP32 can physically plug into the pod at all

If you're short on time, do these four: M18 (free, ten seconds), M17 (protects your ESP32), then M2 and M6.